

Smart Urban Risk & Resource Optimization System (SURROS)

1 Problem Statement (Clear & SIH-Level)

Urban areas me multiple risks hote hain:

- Traffic congestion
- Road accidents
- Pollution hotspots
- Waterlogging
- Emergency response delay

Lekin problem ye hai:

- Data alag-alag departments me scattered hota hai
- Koi integrated predictive system nahi hota
- Resource allocation reactive hota hai (incident ke baad)
- Preventive planning weak hoti hai

Core Problem:

Develop an intelligent web-based system that predicts urban risk levels for different city zones using historical data and optimizes allocation of limited public resources under constraints.

In simple words :

“Primary users: Municipal administrators, traffic control departments, and emergency planning authorities.”

MUST-HAVE FEATURES (Core Functional Features)

1 Zone-wise Risk Prediction

User map ya dropdown se area select kare →

System:

- Current risk score show kare
- Next 7 days ka predicted risk trend graph show kare
- Risk category bataye (Low / Medium / High)

👉 Without this, project ka main soul hi nahi hai.

2 Risk Breakdown Explanation

Sirf number nahi.

System bataye:

- Traffic ka contribution kitna %
- Accident ka contribution kitna %
- Weather ka contribution kitna %

Matlab explainability.

“Why this area is high risk?”

Ye feature bahut powerful lagega judges ko.

3 Resource Allocation Recommendation

System automatically suggest kare:

- Kitne traffic police
- Kitne ambulances
- Monitoring level

Constraints ke andar optimized solution.

Without this → sirf prediction project ban jayega.

4 Data Insights Page (EDA)

Graphs:

- Risk distribution
- Correlation heatmap
- Zone comparison

Ye show karega ki team ne data samjha hai.

5 Seasonal & Monthly Risk Analysis (NEW CORE FEATURE)

System bataye:

- Kis month me sabse zyada risk
- Kis season me spike
- Trend graph (Jan–Dec)
- Year-over-year comparison

Ye tumhara strong insight feature hai.

6 Admin Panel

- Data upload kar sake
- Model retrain kar sake
- New zone add kar sake

SIH me ye feature impress karta hai.

Bonus Features (If Team Strong Hai)

- Alert system (email / notification)
- Historical trend replay mode
- Export report as PDF
- Risk map heat visualization

